PROVISIONAL PATENT APPLICATION

Under Section 9 of the Patents Act, 1970

TITLE OF THE INVENTION

TRACELN: AN ARTIFICIAL INTELLIGENCE POWERED UNIFIED CAMPUS ACADEMIC OPERATIONS PLATFORM INTEGRATING MULTI-MODAL BIOMETRIC ATTENDANCE VERIFICATION, ATTENDANCE-GATED ADAPTIVE CONTENT DELIVERY, REAL-TIME LIVE SESSION INTELLIGENCE, AND CROSS-SESSION STUDENT KNOWLEDGE GRAPH MANAGEMENT FOR INSTITUTIONS OF HIGHER EDUCATION

1. FIELD OF THE INVENTION

The present invention relates to the field of educational technology systems and artificial intelligence applications in academic institution management. Specifically, the invention pertains to a unified software platform that integrates:

- multi-modal biometric and digital attendance verification systems,
- artificial intelligence powered adaptive learning content delivery with attendance-based access control,
- real-time live session intelligence with AI-assisted teaching support,
- persistent cross-session student academic knowledge graph management, and
- institutional analytics and communication systems,

for use in colleges, universities, and institutions of higher education, particularly addressing the operational and pedagogical requirements of the Indian higher education ecosystem.

2. BACKGROUND OF THE INVENTION

2.1 The Attendance Problem

In conventional educational institutions, attendance management relies on manual processes including verbal roll calls and paper registers. These methods consume between 8 and 15 minutes of every class session, directly reducing teaching time. Studies across Indian engineering colleges indicate that approximately 12% of total class time is spent on administrative attendance activities rather than instructional content.

Existing digital attendance solutions including QR code scanning systems and biometric terminals address only the recording function without integrating attendance data into any downstream academic workflow. A student marked present in such systems gains no additional academic benefit, and an absent student faces no structured consequence with respect to learning material access.

2.2 The Proxy Attendance Problem

Proxy attendance, wherein one student marks attendance on behalf of an absent peer, represents a significant challenge in Indian higher education. Existing QR-based systems are vulnerable to screenshot sharing. Existing facial recognition systems lack multi-layer verification combining liveness detection, session heartbeat monitoring, and response rate analysis. No existing system employs the combination of biometric verification and periodic liveness challenges within a live digital session to prevent proxy.

2.3 The Content Distribution Problem

Educational content distribution in Indian colleges predominantly occurs through general consumer messaging applications such as WhatsApp and Telegram. This approach lacks access control, version management, comprehension verification, and download tracking. Content shared through such channels is frequently redistributed outside the institution, misattributed, or lost in message history. No mechanism exists to link content access permissions to the student's attendance status or comprehension level.

2.4 The Live Class Intelligence Gap

During live teaching sessions conducted via video conferencing platforms such as Zoom, Google Meet, and Microsoft Teams, instructors have no mechanism to monitor individual student comprehension in real time, detect widespread confusion before it becomes a learning deficit, receive AI-generated teaching suggestions without disrupting the session flow, or maintain a real-time record of student engagement patterns. Existing platforms provide participant count and basic engagement metrics but do not process session audio for comprehension intelligence or provide actionable instructor interventions.

2.5 The Student Knowledge Continuity Problem

Academic performance data in existing systems is fragmented across examination results, attendance records, and informal teacher observations. No existing system maintains a persistent per-student topic-level mastery profile that is automatically updated from live session events and used to personalise the preparation experience for subsequent sessions. Teachers begin each class without structured knowledge of which specific topics each specific student understood or struggled with in previous sessions.

2.6 The Bandwidth Inequality Problem

In India, significant portions of the student population in tier-2 and tier-3 cities and rural areas experience internet connectivity below 1 Mbps. Existing live learning platforms provide no mechanism to maintain educational continuity and attendance validity for students experiencing poor connectivity. Students on poor connections either miss class entirely or receive degraded audio-only experience without any structured content support.

2.7 The Institutional Visibility Problem

Department heads, principals, and academic administrators in Indian colleges rely on manually compiled weekly or monthly reports to assess academic performance. Real-time visibility into class-level engagement, department-level attendance trends, and student-level risk indicators is not available in any existing integrated platform. Decision-making occurs on stale data, resulting in delayed interventions for at-risk students.

3. OBJECTS OF THE INVENTION

The primary objects of the present invention are as follows:

1. To provide a unified platform eliminating manual attendance management in educational institutions through multi-modal automated attendance verification.
2. To create a direct operational linkage between student attendance status and learning content access permissions through four configurable access tiers.
3. To prevent proxy attendance through multi-layer verification combining biometric identity, session heartbeat monitoring, liveness challenges, and response rate analysis.
4. To enable AI-powered automatic processing of educational content for summary generation, comprehension quiz creation, and student query response without instructor effort after initial upload.
5. To provide identity-linked watermarked content downloads enabling leak source attribution.
6. To deliver real-time AI session intelligence to instructors during live teaching sessions through a configurable three-mode AI assistant.
7. To detect student confusion, engagement decline, pace anomalies, and energy reduction during live sessions and deliver actionable instructor interventions.
8. To maintain persistent per-student topic mastery knowledge graphs updated automatically after every session.
9. To generate personalised pre-class briefings for instructors and personalised warmup content for individual students from knowledge graph data.
10. To maintain attendance validity and learning continuity for students experiencing poor internet connectivity through adaptive bandwidth mode.
11. To provide real-time institutional analytics to department heads and principals without manual report compilation.

12. To automate parent and tutor notifications triggered by attendance and performance thresholds.
13. To serve five distinct institutional roles — student, teacher, tutor, head of department, and principal — through a unified database architecture.

4. SUMMARY OF THE INVENTION

The present invention TRACELN provides a computer-implemented unified campus academic operations platform comprising thirteen integrated modules operating against a single shared database.

The platform accepts input from student mobile devices, teacher workstations, and administrator consoles. It processes biometric data, session audio transcripts, educational content files, and institutional configuration parameters.

The system produces attendance records, gated content access decisions, real-time AI observations, session health reports, student knowledge graph updates, institutional analytics, and automated stakeholder communications as outputs.

The novel contribution of the invention lies in the integration of attendance verification as a dynamic prerequisite for content access, combined with live session AI intelligence and cross-session student knowledge persistence, within a single unified platform serving all institutional roles simultaneously.

5. DETAILED DESCRIPTION OF THE INVENTION

MODULE 1 — Multi-Modal Attendance Subsystem

The attendance subsystem comprises three verification modalities operating against a single unified AttendanceSession database table, ensuring that regardless of verification method used, downstream content access control and analytics receive identical data structures.

Modality 1A — QR Code Session Attendance

The instructor initiates an attendance session through the teacher interface. The system generates a time-limited QR code with configurable validity window of 1 to 30 minutes. The QR code encodes a session token, institution identifier, subject identifier, and cryptographic hash preventing duplication.

Students scan the QR code using the TRACELN mobile application. The application submits the scanned token along with device identifier and GPS coordinates to the attendance API endpoint. The system validates token authenticity, checks for duplicate submissions from the same device, validates the student's enrollment in the subject, and creates an attendance record on successful validation.

Modality 1B — Facial Recognition Attendance

The system maintains a face embedding database constructed from student enrollment photographs. During attendance sessions, the student presents their face to the device camera. The application captures a face image, extracts facial landmarks, computes a face embedding vector, and transmits the vector to the verification API.

The API computes cosine similarity between the submitted embedding and the enrolled embedding for the claimed student identity. Similarity above a configurable threshold (default 0.85) confirms identity. Liveness detection through blink detection and head movement challenge prevents photograph spoofing.

Modality 1C — Live Session Heartbeat Attendance

During live online sessions, the student client transmits a heartbeat signal to the session server via persistent WebSocket connection at 120-second intervals. Each heartbeat includes session token, participant identifier, and timestamp.

The system counts valid heartbeats received per participant. Attendance is marked valid when heartbeat count exceeds a minimum threshold calculated as 60% of expected heartbeats for the session duration.

Join-and-leave detection monitors WebSocket connection events. Participants joining and disconnecting within the first 10 minutes are flagged for manual review.

Periodic liveness challenges transmit a randomized button interaction request to the student interface. Failure to respond within 45 seconds is recorded as a liveness failure event and reduces effective heartbeat count.

Response rate monitoring calculates the ratio of pulse check responses to pulse checks sent per participant. Participants with response rates below 20% over three consecutive pulse checks are flagged as potentially inactive.

MODULE 2 — Attendance Dispute Resolution System

Students may submit attendance dispute requests through the student interface within a configurable window (default 48 hours) after session closure.

Each dispute submission captures the student identifier, session identifier, claimed attendance status, reason category selection, optional supporting description, and submission timestamp.

Disputes are routed to the subject teacher for review. The teacher interface displays session logs, student connection records, and heartbeat data alongside the dispute claim. The teacher approves or rejects with mandatory reason documentation.

Approved disputes update the attendance record and cascade to content access recalculation. All dispute actions are logged with actor identifier and timestamp for audit purposes.

MODULE 3 — Attendance-Gated Content Capsule System

Educational content is organised into units called capsules. Each capsule contains one or more content items including PDF documents, video files, and supplementary materials.

Capsule Access Control

Access mode is set per capsule by the instructor at creation time and may be modified at any time. Four access modes are defined:

- Access Mode 1 — Open: No attendance prerequisite. All enrolled students may access content at any time.
- Access Mode 2 — Session Active: Content accessible only to students whose attendance is confirmed for the current active session. Accessible only during session window.
- Access Mode 3 — Post Attendance Mark: Content accessible after the instructor marks the session attendance as finalised. Students present in the session receive access. Absent students receive restricted access or no access per instructor configuration.
- Access Mode 4 — Attendance Percentage Gate: Content accessible only to students whose cumulative attendance percentage in the subject meets or exceeds a configurable threshold (e.g., 75%). Percentage is calculated dynamically from attendance records at time of each access request.

Content Access Evaluation

Every content access request triggers a real-time evaluation sequence: (1) retrieve the student's attendance record for the current session; (2) retrieve the student's cumulative attendance percentage for the subject; (3) evaluate against the capsule access mode; (4) grant or deny access; (5) log the access decision with timestamp. Access denials return a structured message indicating the specific requirement not met and the student's current status toward meeting it.

MODULE 4 — AI Content Processing Pipeline

Upon capsule content upload, the AI processing pipeline is triggered automatically through five sequential stages:

Stage 1 — Content Extraction

PDF documents are parsed to extract text content. Images within PDFs are processed through optical character recognition where text is present. Video files are processed through speech-to-text transcription.

Stage 2 — AI Summary Generation

Extracted text is submitted to a large language model API with a structured prompt requesting a 200–300 word educational summary capturing key concepts, definitions, and relationships.

Stage 3 — Comprehension Quiz Generation

The same text is submitted with a prompt requesting generation of five multiple choice questions with four options each, one correct answer, and a brief explanation per question. Questions are stored in a structured format linked to the capsule.

Stage 4 — FAQ Index Construction

Key concepts and likely student queries are extracted and indexed. When a student asks a question about the capsule content, the system retrieves relevant passages and generates a contextual answer.

Stage 5 — Download Gate Configuration

If the capsule is configured with comprehension gating, students must achieve a minimum quiz score (configurable, default 60%) before the download button becomes active.

Watermarked Download

When a student initiates a content download, the following sequence executes: (1) student identity is verified against session token; (2) access permission is evaluated; (3) quiz completion status is checked if comprehension gating is active; (4) a unique download instance is created in the database; (5) the PDF is processed to embed an invisible watermark containing student full name, roll number, institution name, download timestamp, and unique download instance identifier; (6) a signed URL is generated with configurable expiry (default 10 minutes); (7) the download event is logged; (8) the signed URL is returned to the client.

MODULE 5 — Three-Mode AI Session Brain

The AI Session Brain is the core intelligence module activated during live teaching sessions. It operates in one of three configurable modes selectable by the instructor before or during the session.

Mode 1 — Silent Observer
The AI Session Brain receives audio transcript chunks from the live session every 120 seconds. It processes each chunk through the observation generation pipeline and stores observations in the session observation database table. Observations are displayed only in the instructor's sidebar interface. Students are not informed that AI observation is active. The AI does not communicate with students in this mode. The pre-session brief is generated in this mode based on previous session data.

Mode 2 — Active Assistant
In addition to silent observation, the AI generates intervention suggestions when observation confidence exceeds a threshold. Suggestions are displayed to the instructor as actionable prompts with approve or dismiss options. The AI does not take any action without explicit instructor approval. Approved actions may include sending a recap message to students, launching a pulse check, or posting a resource to the class wall.

Mode 3 — Autonomous Teaching Assistant
The AI actively participates in student interaction within instructor-defined boundaries. It monitors the class doubt wall and responds to student questions within its knowledge scope. Queries outside its defined scope are escalated to the instructor with the student's question and context.

The AI generates and dispatches pulse checks autonomously when comprehension indicators fall below threshold without waiting for instructor approval. The instructor retains ability to override or dismiss any autonomous action.

MODULE 6 — Five-Type AI Intervention System

The intervention system evaluates session state every 60 seconds and generates at most one intervention per evaluation cycle. A minimum 7-minute cooldown period is enforced between successive interventions to prevent instructor disruption.

Intervention Type 1 — Confusion Alert
Trigger condition: The count of students who have not responded to any interaction in the preceding 8 minutes meets or exceeds 4 AND represents more than 25% of total session participants. The intervention displays the number of silent students and duration of silence, with options to send recap, launch pulse check, or dismiss.

Intervention Type 2 — Pace Alert
Trigger condition: Time elapsed since last pulse check exceeds 20 minutes AND total session duration exceeds 25 minutes. The intervention provides a recommendation to run a quick comprehension check, with options to send pulse check, continue, or dismiss.

Intervention Type 3 — Concept Bridge
Trigger condition: Current transcript contains a topic that has recorded weak mastery scores in the knowledge graph for 3 or more session participants. The intervention identifies the topic name and number of students with weak prerequisite mastery, recommending dispatch of a refresher resource.

Intervention Type 4 — Energy Check
Trigger condition: Session duration exceeds 45 minutes AND current elapsed minutes modulo 45 is less than 6. The intervention acknowledges natural attention decline patterns and recommends a brief break.

Intervention Type 5 — Topic Completion
Trigger condition: AI analysis of transcript detects a topic transition signal indicating the instructor has concluded a topic section. The intervention identifies the completed topic and recommends a quick comprehension check, with options to launch pulse check, continue, or dismiss.

MODULE 7 — Live Pulse Check System

A pulse check is a single multiple choice question delivered to all session participants simultaneously.

Creation
The instructor may create a pulse check manually by entering question text and four answer options, or may request AI generation. When AI generation is requested, the system submits recent session transcript to the AI model with a prompt requesting a comprehension question grounded in the last 5 minutes of teaching content.

Delivery
The pulse check is transmitted to all active session participants via WebSocket with a configurable response duration (default 30 seconds). A countdown timer is displayed on student interfaces. Students select one answer option.

Collection and Analysis
Responses are collected in real time. The instructor interface displays a live response count and distribution across options. After closure, the system calculates total responses received, correct response count, correct response percentage, and per-option selection count and percentage.

AI insight generation submits the distribution data to the AI model which generates a natural language observation about what the distribution reveals about class comprehension, including specific patterns in incorrect answer selections. The pulse check result and AI insight are stored in the session record and included in the post-session health report.

MODULE 8 — Anonymous Live Doubt Wall

The class wall provides a communication channel for anonymous student question submission during live sessions.

Submission

Students submit doubt text through the student interface. The submission is stored with the student's identity in the database but displayed to other students without any identifying information. Only the instructor interface displays student identity alongside doubt text.

Resonance Mechanism

Other students may indicate shared confusion by selecting a resonance indicator on any existing doubt. Each selection increments the resonance count for that doubt.

Hot Doubt Escalation

When a doubt's resonance count reaches or exceeds a configurable threshold (default 3 additional students), the doubt is classified as a hot doubt. The following automated actions occur: (1) the instructor receives a real-time push notification identifying the hot doubt; (2) the doubt is visually highlighted in the instructor interface; (3) a WebSocket event is transmitted to the AI Session Brain for possible intervention consideration.

Doubt Resolution

The instructor may mark a doubt as addressed. Addressed doubts are automatically included in the post-session capsule under a Doubts Addressed section.

MODULE 9 — Session Health Report System

Upon session closure, the system automatically computes a comprehensive session health report comprising the following metrics:

- Attendance Score: Calculated as the ratio of students who attended the minimum required duration to total enrolled students, expressed as a percentage normalised to a 0–100 score.
- Engagement Score: Computed from the weighted combination of pulse check participation rate, doubt wall activity, and heartbeat response consistency.
- Comprehension Score: Computed from average pulse check correct response percentage across all pulse checks conducted in the session. If no pulse checks were conducted, the field is marked as unmeasured.

- Pace Score: Computed from the ratio of topics covered to time elapsed compared to historical session pace data for the same subject.
- Overall Health Score: Weighted average of all four component scores with configurable weights (default: attendance 25%, engagement 25%, comprehension 35%, pace 15%).

AI narrative generation submits the computed metrics to the AI model requesting a 2–3 sentence narrative observation identifying the most significant insight and one specific actionable recommendation for the next session. The complete health report is stored, displayed to the teacher immediately after session closure, included in HOD analytics aggregation, and used as input for the next session's pre-class brief generation.

MODULE 10 — Cross-Session Student Knowledge Graph

The student knowledge graph maintains a persistent record of per-student per-topic academic mastery across all sessions.

Knowledge Graph Structure

Each record contains: student identifier, subject identifier, topic name, mastery percentage (0–100), understanding level classification (strong / moderate / weak / not covered), confidence score, sessions seen count, times confused count, times demonstrated understanding count, last assessed session identifier, and last updated timestamp.

Update Mechanism

After every live session closure, the update pipeline runs for each participant through five steps:

- Step 1: Retrieve all pulse check responses for the participant in the session and calculate correct response percentage.
- Step 2: Retrieve all AI observation events from the session where the participant is in the affected student list. Count confusion events and positive understanding events per topic.
- Step 3: For each topic with signals, apply the mastery update formula: new_mastery = (old_mastery × 0.6) + (session_score × 0.4), where session_score is derived from correct response percentage adjusted by confusion and understanding event counts.
- Step 4: Update understanding level classification based on new mastery percentage thresholds: 75% and above = Strong; 50% to 74% = Moderate; 1% to 49% = Weak; 0% = Not covered.
- Step 5: Commit all knowledge graph updates to database.

Pre-Class Brief Generation

30 minutes before each scheduled session, the brief generation pipeline executes: (1) query knowledge graph for all students enrolled in subject and retrieve weak and moderate understanding records; (2) query last session health report for same teacher and subject; (3) query recent failed capsule quiz attempts for the subject; (4) submit aggregated data to the AI model requesting class readiness score, list of students needing attention, topic recommended for revision, warmup activity suggestion, and predicted difficult concepts; (5) deliver brief to teacher interface upon session start.

Personalised Student Warmup

Simultaneously, individual warmup content is generated per student. Students with strong mastery on prerequisite topics receive a preview challenge question for the new topic. Students with weak mastery on prerequisite topics receive a targeted micro-revision capsule covering their specific weak topics. Students who were absent in the previous session receive an AI-generated summary of what they missed with key points and important concepts.

MODULE 11 — Adaptive Bandwidth Mode

The adaptive bandwidth module monitors connection quality for every session participant continuously throughout the session.

Monitoring

WebRTC connection statistics are sampled every 30 seconds including available downlink bandwidth estimate, packet loss percentage, round trip time, and video frame decode rate.

Threshold Detection

When available bandwidth falls below 1 Mbps for two consecutive sampling intervals, the affected participant's session mode is switched to low-bandwidth mode.

Low Bandwidth Mode Operation

Upon detection of low bandwidth, the following sequence is executed: (1) the student's video subscription is suspended to reduce bandwidth consumption; (2) a notification is sent to the student interface explaining the mode switch; (3) the teacher interface receives a bandwidth alert identifying the affected student; (4) the AI low-bandwidth pipeline activates for that participant — every 2 minutes, the most recent session transcript chunk is submitted to the AI model requesting a 3–4 sentence plain text summary of what was just taught; (5) the summary is delivered to the student via a lightweight WebSocket text message; (6) the student's heartbeat signal continues operating normally, maintaining attendance validity; (7) when connection quality recovers above threshold, video mode is automatically restored.

Full Capsule Preservation

Regardless of which mode the student operated in during the session, they receive full access to the post-session capsule if their attendance record is valid. Bandwidth degradation does not disadvantage the student's post-session learning access.

MODULE 12 — Institutional Analytics System

The analytics system aggregates data from all platform modules into role-specific dashboard views with real-time refresh.

Teacher Dashboard

Displays subject-wise attendance trend charts, student-wise attendance percentage rankings, capsule engagement metrics (views, quiz attempts, downloads), session health score history, at-risk student identification, and knowledge graph summary per subject.

Head of Department Dashboard

Displays department-wide attendance trends across all faculty and subjects, faculty-wise session activity and capsule creation metrics, subject-wise comprehension trends from pulse check aggregation, student risk distribution, capsule content coverage by subject, and live session frequency and health score averages by faculty.

Principal Dashboard

Displays institution-wide attendance trends, department comparison metrics, student population risk analysis, faculty activity overview, and system usage statistics. All dashboard data refreshes in real time without requiring manual report generation or data compilation.

MODULE 13 — Automated Communication System

The communication module integrates with WhatsApp Business API to deliver automated notifications triggered by platform events.

Notification Triggers

- Trigger 1 — Student Absence (Single): Recipient: subject teacher and tutor. Condition: student absent from a session. Content: student name, subject, session date, current attendance percentage.
- Trigger 2 — Consecutive Absence: Recipient: tutor. Condition: student absent from 2 consecutive sessions in same subject. Content: student name, sessions missed, attendance percentage, subjects affected.

- Trigger 3 — Critical Attendance: Recipient: parent or guardian. Condition: student cumulative attendance falls below 75% in any subject. Content: student name, subject names, attendance percentages, advisory message.
- Trigger 4 — Quiz Failure: Recipient: tutor. Condition: student fails capsule quiz below passing threshold twice for same capsule. Content: student name, capsule title, attempt scores.

All notification content, recipient lists, and trigger thresholds are configurable by institution administrators. Notification delivery uses existing WhatsApp Business API integration without requiring separate messaging infrastructure.

6. PROVISIONAL CLAIMS

Note: Full detailed claims will be provided in the complete specification filed within 12 months of this provisional application. The following provisional claims establish the scope of the invention:

14. A computer-implemented unified campus academic operations platform comprising multi-modal attendance verification, attendance-gated content delivery, AI-powered live session intelligence, and cross-session student knowledge graph management operating against a single unified database.
15. The platform of claim 1, wherein attendance verification operates through QR code, facial recognition, and live session heartbeat modalities.
16. The platform of claim 1, wherein learning content access is dynamically determined by attendance status across four configurable access tiers.
17. The platform of claim 1, wherein an AI engine processes uploaded educational content to generate summaries, comprehension quizzes, and query responses automatically.
18. The platform of claim 1, wherein downloaded content is watermarked with student identity information and delivered via expiring signed URLs.
19. The platform of claim 1, wherein a three-mode AI teaching assistant monitors live sessions and delivers real-time observations and interventions to the instructor.
20. The platform of claim 6, wherein five categories of instructor interventions are generated with configurable triggering conditions and a minimum cooldown period between successive interventions.

21. The platform of claim 1, wherein anonymous student doubt submissions are aggregated with resonance counting and automatically escalated to the instructor upon exceeding a threshold.
22. The platform of claim 1, wherein a persistent per-student topic mastery knowledge graph is updated automatically after each session and used to generate personalised pre-class preparation content.
23. The platform of claim 1, wherein upon bandwidth degradation below a threshold, AI-generated text summaries are delivered to the affected student while maintaining attendance validity.
24. The platform of claim 1, serving five distinct institutional roles through role-specific dashboard views aggregating real-time data from all platform modules.

7. ABSTRACT

The present invention relates to TRACELN, a computer-implemented unified campus academic operations platform for institutions of higher education. The platform integrates multi-modal attendance verification through QR code scanning, facial recognition, and live session heartbeat monitoring, with all modalities sharing a unified database. Student attendance status directly determines access to educational content capsules through four configurable access tiers. Uploaded content is automatically processed by artificial intelligence to generate summaries, comprehension quizzes, and student query responses. Downloads are individually watermarked with student identity for leak attribution.

During live sessions, a three-mode AI teaching assistant monitors session transcripts and delivers real-time natural-language observations and five categories of instructor interventions. Anonymous student doubt submissions are aggregated with resonance-based hot doubt escalation to the instructor. Post-session health reports and per-student knowledge graphs are generated and updated automatically.

Students experiencing poor connectivity receive AI-generated text summaries with attendance validity maintained. Institutional analytics serve five roles in real time. Parent and tutor notifications are automated through WhatsApp Business API integration.

The invention is novel in integrating all described functions within a single unified platform specifically designed for the Indian higher education ecosystem.